

PrimusCredence

Provable Security

primuscredence.com

UAE

Dr. Raghavendra Ramesh

Founder & CEO, PrimusCredence

22+ YEARS OF EXPERIENCE IN PROVABLE SECURITY

PhD, IISc Bangalore

Model checking for information-flow security

Oracle Labs · 2014–2019

Java vulnerability detection

ConsenSys R&D · 2019–2021

Cross-chain protocols, consensus verification

Supra · VP of R&D · 2021–ongoing

BFT design & FV, bridges, DeFi

TECHNIQUES

Static analysis, theorem proving,
model checking

TOOLS

Dafny, TLA+, IVy, Lean, Isabelle

Where the Risk Actually Lives

The cloud base is proven. The app above it — and the seams between — are not.

Third-party plugins & extensions

Run inside the app — and inherit its privilege

APPLICATION

Its own roles, API tokens and public users — the authorisation logic and configuration in its database and admin panel

OWASP #1 Broken Access Control · #2 Misconfiguration — in 100% of apps

THE APP ↔ CLOUD & LIBRARY SEAM

- Access control via APIs
- SDK & library usage
- PQC app↔cloud binding

CLOUD & INFRASTRUCTURE

VMs · containers · databases · queues · object storage · IAM · network

Access control, network & PQC transport — proven: Zelkova, Tiros, ML-KEM

UNPROVEN TODAY

Where the breaches are

The app's own access control, its plugins, and the API, library and PQC seams beneath it.

→ **This is what we prove.**

PROVEN BASE

Cloud IAM, network & PQC transport — already formally verified.

OWASP 2025 — Access Control & Misconfiguration

The two categories at the top are found in 100% of apps tested.



Broken Access Control

#1 — unchanged

40 CWEs · 1.84M occurrences · 32,654 CVEs
SSRF now folded in — easy to exploit, widespread.



Security Misconfiguration

#2 — up from #5

16 CWEs · 719,084 occurrences
Again, 100% of tested apps carried a form of it.

BROKEN ACCESS CONTROL IN THE WILD

Capital One	106M	WAF SSRF → over-privileged role; a tipster found it, ~4 months in
Optus	9.5M	access-control bug on a dormant, internet-facing API — latent ~4 years
T-Mobile	37M	API data retrieval without authorization — ~6 weeks to notice
Dell	49M	auto-approved fake partner, then service-tag enumeration
Trello	15M	unauthenticated email lookup through rotating proxies
Twilio Authy	33M	unauthenticated endpoint enumerated for phone numbers

Every one was a well-formed, authenticated request a monitor read as normal.

Apps Misuse the APIs They Run On

Cloud and app APIs are protocols, not calls — misuse still compiles.

THE PATTERN

- An object fetched by request ID with no ownership check — BOLA / IDOR
- A credential written to a log; pagination left undrained
- Authorisation in middleware a scanner can't see

THE TREND IS NOT REVERSING

- +29% IDOR reports year on year (HackerOne 2025)
- +36% access-control criticals — now #1 (Bugcrowd 2025)
- ~95% of API attacks carry a valid token — authentic, not authorised

\$4.4M

average cost of a data breach
(IBM 2025)

241 days

mean time to identify and
contain (IBM 2025)

258 / day

API attacks per enterprise —
2× 2024 (Akamai 2025)

A correct policy invoked through a misused SDK still leaks.

The Base Went Post-Quantum. Are the Apps Ready?

The internet and cloud layer are shifting to hybrid PQC — and it is proven.

ALREADY SHIFTING, ALREADY PROVEN

- NIST FIPS 203 / 204 / 205 (Aug 2024)
- Hybrid ML-KEM default in browsers
- Live at Cloudflare, AWS, Google, Apple
- ML-KEM itself machine-checked (2024)

THE APP ON TOP IS NOT

- No identity binding → impersonation
- Downgrade to legacy → decrypt later
- KEM key reuse → forward secrecy lost
- Token not channel-bound → replayed

LONG-SECRET VERTICALS BITE HARDEST

Gov ID (10+ yr) · PKI · healthcare · defence (25–50 yr).
A mis-bound 10-year credential is a decade-long liability.

A proof at the transport layer says nothing about the app on top of it.

AI Industrialises the Attacker

A sampling defence cannot answer an enumerating attack.

4.5×

revenue of AI-assisted scam operations versus non-AI equivalents

\$3.52

average compute cost per LLM-agent exploit attempt on a real CVE

87%

of a 15-CVE sample exploited from the public description alone

TRADITIONAL PEN TEST

10^2 – 10^3 paths, human-selected
2–4 week window, then closed
High marginal cost per path
“We found no way in”

AI-DRIVEN ADVERSARY

10^5 – 10^7 paths, machine-enumerated
Continuous, no window
Near-zero marginal cost
Approaching exhaustive

Proof is the one capability that answers exhaustive search in kind.

Why Today's Security Stack Isn't Enough

All essential, all stay — but each only samples what happened.

SAST / DAST / IAST

SCA / supply chain

ASPM

WAF / API security

CSPM · CIEM · CNAPP

Pen test & red team

SOC · EDR · SIEM

Policy-as-code

- **SAST** Knows code patterns, not intent. A check present but wrong still passes.
- **DAST** Only crawled endpoints and seeded data; even multi-persona only samples.
- **IAST / RASP** Same-code-path BOLA is invisible — was the call even permitted?
- **SCA / ASPM** Known CVEs and risk diffs; never whether a grant is actually correct.
- **CNAPP / CSPM** Stops at the cloud IAM role; never reads the app's own roles.
- **WAF / API sec** A permitted request and a legitimate one look identical on the wire.

They all share one gap: no statement of what is allowed, decided over every request.

Keep every one of them — and add the layer that closes that gap.

Proof — Now Feasible

Formal verification used to take expert teams years. AI brings it into range.

THEN

- Expert teams, bespoke specs, years
- Confined to avionics, silicon, protocols
- “End users won’t write a spec”

NOW — AI-ASSISTED

- AI assists modelling & proof search
- Expert owns & ratifies the spec
- AWS runs ~1B SMT queries a day

What we add A preventive control with evidence behind it, beside your testing and SOC.
We add to your stack; we don’t replace it. Proof alone isn’t enough.

WHAT STAYS WITH YOU — ALL OF IT NEEDED

Configuration drift, shadow assets and anything outside the model

Insider misuse, and valid credentials used exactly as intended

Detection, response and the rest of the programme you already run

Automated Reasoning

Proof establishes that bad things cannot happen.

1

Formalise

Your existing artefact becomes a logical model

2

State

The bad state you fear becomes a formula

3

Solve

Z3, Batfish, IVy, TLA+ decide it; CBMC to a bound

4

Report

A proof, or a reproducible attack path

✦ AI - POWERED ✦

UNSAT

No violating state exists — over the entire space, under a stated model.

Not “none was found”.

SAT + witness

The exact request, path or call ordering that breaks it — reproducible on your estate.

Industrial practice: AWS (Zelkova, Tiros, Nitro) · Microsoft (Z3) · Meta (Infer) · Airbus (Astrée)

Solutions

1 App entitlement proofs

Who can reach what?

2 App→cloud escalation

A path through the app?

3 App & API conformance

SDKs used correctly?

4 Post-quantum migration

Bound on a PQC base?

5 Agentic guardrail verification

What an agent may do?

1. Application Access-Policy & Entitlement Verification

What can each app role, API token and public user do inside the app — and can any of them reach what they should not?

THE PROBLEM

An app's authorisation lives in its own database and admin panel — which role reads which type, what the Public role exposes. No SAST, CSPM or CNAPP reads it; DAST only samples.

Portable: it reads the app's own config, so it holds on-prem, not only in cloud.

PRECEDENT Cedar + SymCC → Cedar Analysis
BUYERS Platform & AppSec teams · ISVs

WHAT WE BUILD

- The app's roles, API tokens and public users, into Cedar
- SymCC decides over all requests — BOLA, BFLA, BOPLA
- CheckNoNewAccess gates every role change in CI

OUTPUT

Proof that no role, token or public user reaches what it must not — over all requests — or the exact request that breaks it.

2. App-to-Cloud Escalation-Path Proofs

Can a non-admin user reach the app's cloud role — its secrets, buckets and keys — through a plugin or an IdP group?

THE PROBLEM

Each layer passes on its own — the cloud role is not public, the plugin is legitimate, every call looks normal. No tool composes them: CNAPP starts from a vulnerable workload, not an app role.

On-prem only the far end changes — the path through the app is identical.

PRECEDENT AWS Zelkova + Tiros, composed
BUYERS AppSec teams · plugin-heavy app stacks

WHAT WE BUILD

- Plugins and IdP groups as principals, with their grants
- App role → plugin → cloud IAM role → data, one graph
- Admin-panel and network reachability as the boundary

OUTPUT

The path through the app, edge by edge — each step checkable in the layer that owns it — or a proof no such path exists.

3. Application and API-Usage Conformance

Does our own code use the APIs it is built on — cloud, credential and crypto SDKs, and its own data-access calls — the way their contracts

THE PROBLEM

Cloud and app APIs are protocols, not calls. Undrained pagination, a logged credential, an object fetched by request ID without an ownership check — all compile and pass review. Same risk

A correct policy invoked through a misused SDK still leaks. Bounded by the library.

PRECEDENT AWS RAPID → CodeGuru · 76% accepted
BUYERS Platform and AppSec teams · ISVs

WHAT WE BUILD

- Object-ownership (BOLA/IDOR) and auth-routing patterns
- A conformance gate for library publishers
- PR-gate CI on Soot, CodeQL and Infer

OUTPUT

The call site, the taint trace, the pattern violated and the corrected call sequence — plus which patterns were checked.

4. Post-Quantum Cryptography Migration

On an already quantum-safe cloud, does the app's integration still authenticate the right party in every session?

THE PROBLEM

The cloud's PQC transport can be formally proven and the app on top still relayed or downgraded. Hybrid migration adds identity-binding and combiner failures that code review ships past.

Verified app crypto on an already-verified cloud — proof end to end.

PRECEDENT PQ-SSH counterexample, then proved (2024)

BUYERS Gov ID · PKI · banks · healthcare · defence

WHAT WE BUILD

- Discovery and a versioned CBOM, gaps listed
- The app-to-cloud integration protocol, transport proof discharged
- Identity, channel and downgrade binding, over all sessions

OUTPUT

The CBOM and a sequenced plan; then a proof the app's integration preserves authentication and secrecy on the verified base — or the relay that breaks it.

AI Agents Multiply Faster Than the Rules

An agent is a non-human identity with delegated privilege and a writable channel.

THE RULES ARE ARRIVING

- EU AI Act — Art. 15 robustness & security
- ISO/IEC 42001 — AI management system
- ISO/IEC 24029-2 — formal methods for AI
- DIFC Reg 10 · Dubai AI Seal · SDAIA

The assurance market does not exist yet — anywhere. You would be early, not late.

THE CHALLENGE — GUARDRAILS ESCAPE

A guard model judging a model is detective, not preventive. The question is not “how often does it fail?” but “can it do this at all?”

-
- 5 What an agent may do**
tool calls, arguments, data access

Our AI-agentic guardrail solution proves the mediation layer, not the model →

5. Agentic AI Guardrail Verification

Not “how often does the agent fail?” but “can it do this at all?”

THE PROBLEM

An agent is a non-human identity with delegated privilege and an instruction channel the adversary can write to. A guard model judging a model is detective, not preventive.

*Provable: no rows outside the caller's ACL.
Not provable: no PII ever reaches the user.*

PRECEDENT Bedrock Automated Reasoning
BUYERS Banks · sovereign AI · OT · healthcare

WHAT WE BUILD

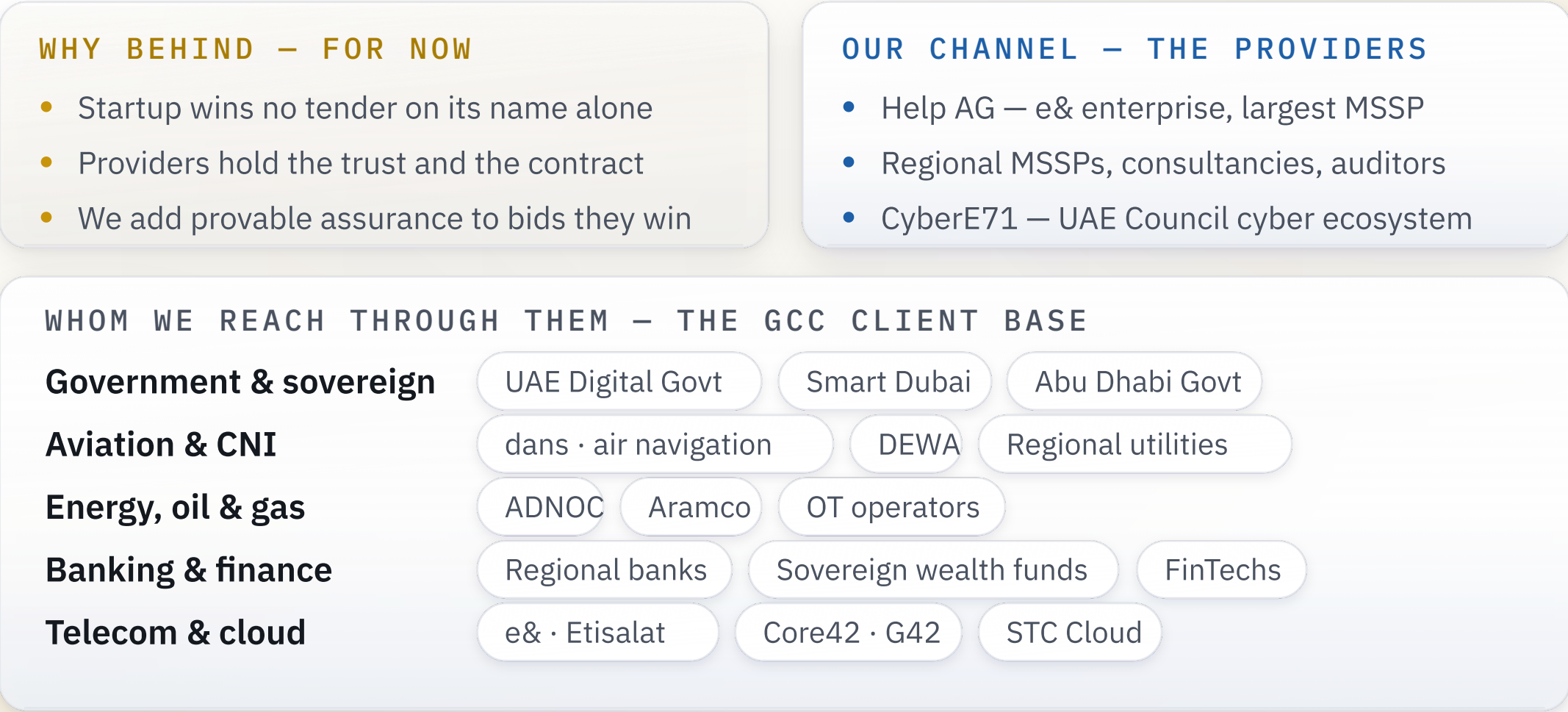
- Tool-call constraints over identity and arguments
- Data-access boundaries in the Solution 1 logic
- Composition safety across call sequences

OUTPUT

A result about the mediation layer, not the model: nothing permitted escapes the envelope, and no path bypasses it.

We Sit Behind the Providers

No reputation yet — we add value to the providers, and through them, their clients.



Why the Provider's Clients Will Ask

Four things happened at once in the GCC — and their clients feel all four.

</

Regulation routes the finding to a second reader. It is not the reason to commission it.

What a Provider Can Resell

We do not certify — so an audit firm or an MSSP is a channel, not a rival.

Scanning · posture reporting · monitoring



Provable control effectiveness

THE COMMERCIAL MOVE

- Specialist capability inside bids you already win
- You keep the client relationship and the contract

WHY IT DEPLOYS EASILY

- Read-only access, no agent on any host
- No scanning window, no production traffic

Read-only and agentless, so it clears change advisory in one pass.

We produce the technical evidence. Your assessor keeps the opinion.

One Result, Three Registers

First line

Engineering and security operations

The property, its model and assumptions, and either the proof or a reproducible attack path.

Second line

Risk

Likelihood moved off a qualitative guess, with residual risk and the model's limits stated.

Third line

Internal audit and compliance

A clause-mapped control assertion: objective, status, owner and re-verification date.

✓ Design → operating effectiveness · attaches to your GRC records · assessors probe less

The first register is the product. The second and third make it worth more than it cost.

Take Away

**Mathematical rigour to security —
proof alongside testing, across apps and AI.**

The shift

Attack path discovery is automated, parallel and cheap. A defence that only samples cannot answer one that enumerates.

The addition

We do not replace testing, monitoring or the SOC. We make one class of controls machine checkable.

The honesty

A proof is relative to a model and a specification — both stated. We produce the evidence; we do not certify.

START HERE — MICRO ASSESSMENT OF SOLUTION 1

One narrow, high-risk slice. Read-only access, no agent deployed, weeks not months — and a counter-example on your own systems, or a proof.

Thank You



primuscredence.com

Dr. Raghavendra Ramesh

Founder & CEO, PrimusCredence
PhD, IISc Bangalore

raghavendra@primuscredence.com

+971 58 189 2803
UAE

Provable Security for Apps & AI Agents